Checkpoint C | Version 2 | Solutions

*Each skill earns **Pass** or **Not Yet**. Some questions contain multiple skills. Grade each skill on its own when it's part of a full question. A **Pass** is given for work showing the core skill. A **Not Yet** is given when there's not sufficient evidence of the core skill. Evidence anywhere on the submission is acceptable but evidence in the question area will be used for the initial grading. Students may request a regrade if evidence exists outside the original grading area and was not included in the original grading. Do not infer a method that the work does not support.*

Cauldrons For Days

The Standard Cauldron is a magical tool used widely in the making of various potions with many buyers and many sellers. The demand and supply curves for the Standard Cauldron can be represented by the following relationships, with prices in coins, and quantities in cauldrons.

$$D : P_b = 90 - \frac{1}{2}Q_d \qquad S : P_s = 15 + Q_s$$

The metal for each cauldron is smelted in foundries outside Hogsmeade, and the fumes drift over the village. The fumes cost residents 12 coins per cauldron made, a cost neither buyers nor sellers pay.

C1.1 | Market Failure

The graph shows the market for cauldrons: marginal private cost (MPC), marginal social cost (MSC), and demand (MB). The lettered regions are for the questions.

Market Quantity ($\hat{Q}$): _____

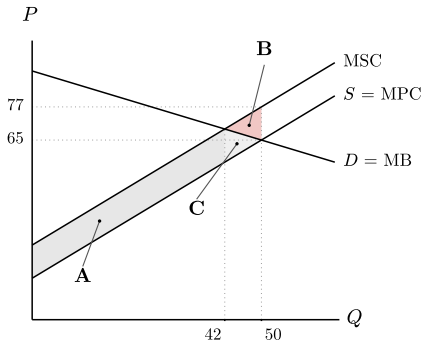
Efficient Quantity (Q^*): _____

Deadweight Loss (*letter*): _____

Deadweight Loss (*area*): _____

For the Last Cauldron Traded: MSC: _____ / MB: _____

The Market Trades: ☐ too many ☐ too few cauldrons



Answers: market quantity **50 cauldrons**; efficient quantity **42 cauldrons**; DWL region **B**; DWL area **48 coins**; last cauldron traded: **MSC = 77**, **MB = 65**; **too many**. The social-cost curve is MSC $= 27 + Q$ the DWL triangle lies between MSC and demand from 42 to 50.

Add the 12-coin external cost to supply. The private and social marginal conditions give:

$$\text{MSC} = 15 + Q + 12 = 27 + Q$$
$$90 - \frac{1}{2}Q = 15 + Q \Rightarrow \hat{Q} = 50, \quad \hat{P} = 65$$
$$90 - \frac{1}{2}Q = 27 + Q \Rightarrow Q^* = 42.$$

At $Q = 50$, MSC is 77 while MB is 65. Trades beyond 42 cost society more than they benefit buyers. The DWL triangle has corners $(42, 69)$, $(50, 77)$, and $(50, 65)$:

$$\text{DWL} = \frac{1}{2} \times 12 \times (50 - 42) = 48 \text{ coins.}$$

Pass (C1.1): Use the supplied graph to identify market and efficient quantities, identify B and calculate its area, and compare MSC with MB at the market quantity to explain overproduction.

A is the external cost on units up to the efficient quantity. C is the private surplus from the excess trades; their external cost is B plus C, so their net social loss is B.

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C2.1 | A Tax for Revenue

Needing revenue, and not yet knowing the size of the externality, the Ministry imposes a tax of 6 coins per cauldron. For this question, ignore the external cost.

Buyer's Price (P_b): _____

Seller's Price (P_s): _____

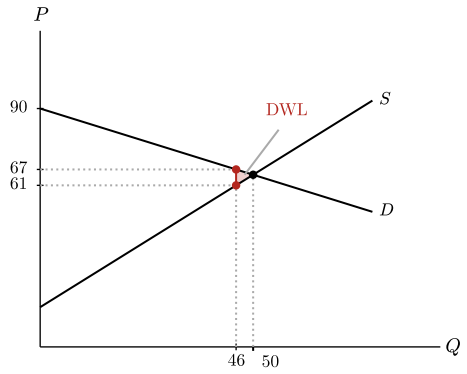
Post-Tax Quantity (Q^{Tax}): _____

Tax Incidence on Buyers: _____

Tax Incidence on Sellers: _____

Deadweight Loss (DWL): _____

Who Bears More of the Tax? ☐ buyers ☐ sellers ☐ equal



Answers: buyer's price **67 coins**; seller's price **61 coins**; post-tax quantity **46 cauldrons**; incidence **2 coins per cauldron on buyers** and **4 on sellers**; DWL **12 coins**; **sellers** bear more.

The tax separates the two prices by 6 coins:

$$P_b = P_s + 6 \Rightarrow 90 - \frac{1}{2}Q = 15 + Q + 6 \Rightarrow Q^{Tax} = 46.$$

Substituting into demand and supply gives $P_b = 67$ and $P_s = 61$. Check: $67 - 61 = 6$. Relative to the untaxed price of 65, buyers pay $67 - 65 = 2$ more and sellers receive $65 - 61 = 4$ less. Thus sellers bear more.

Ignoring the external cost, the efficient quantity is 50. The tax prevents 4 mutually beneficial trades:

$$\text{DWL} = \frac{1}{2} \times 6 \times (50 - 46) = 12 \text{ coins}.$$

Pass (C2.1): Use the tax wedge to find both prices and quantity, calculate each burden against the untaxed price, identify sellers as bearing more, and calculate DWL with the external cost ignored.

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Lovely Lighting from Lumos Lanterns

Many Hogsmeade homeowners hang Lumos Lanterns outside their doors. In addition to providing lighting for the homeowner, each lantern also lights the lane for the neighbors, an external benefit of 4 coins per lantern. Prices are in coins and quantities in lanterns. The demand and supply equations are:

$$D: P_b = 24 - \frac{1}{4}Q_d \qquad S: P_s = 6 + \frac{1}{4}Q_s$$

C1.2 | Market Failure

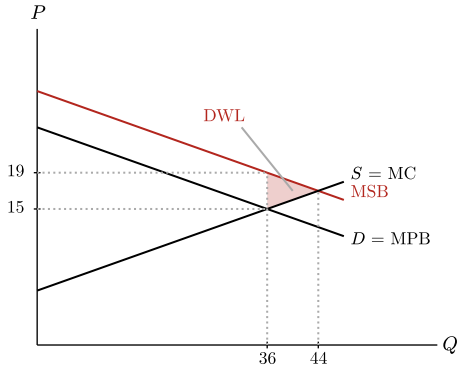
Start by using a graph and algebra to describe the market failure. Graph demand, supply, and MSB, and shade the deadweight loss.

Market Quantity ($\hat{Q}$): _____

Efficient Quantity (Q^*): _____

Deadweight Loss (DWL): _____

The Market Trades: ☐ too many ☐ too few



Answers: market quantity **36 lanterns**; efficient quantity **44 lanterns**; DWL **16 coins**; **too few**. Draw MSB 4 coins above demand (MPB) and shade the triangle between MSB and supply from 36 to 44.

The 4-coin external benefit places MSB above MPB. Market equilibrium uses private benefit; efficiency counts the neighbors' benefit:

$$\text{MSB} = 24 - \frac{Q}{4} + 4 = 28 - \frac{Q}{4}$$
$$24 - \frac{Q}{4} = 6 + \frac{Q}{4} \Rightarrow \hat{Q} = 36, \quad \hat{P} = 15$$
$$28 - \frac{Q}{4} = 6 + \frac{Q}{4} \Rightarrow Q^* = 44.$$

At the last lantern traded, $\text{MSB} = 28 - 36/4 = 19$ and $\text{MC} = 6 + 36/4 = 15$. Trades between 36 and 44 would benefit society more than they cost, but private buyers do not purchase them. The triangle represents their lost surplus:

$$\text{DWL} = \frac{1}{2} \times 4 \times (44 - 36) = 16 \text{ coins.}$$

Pass (C1.2): Draw demand, supply, and MSB; calculate the market and efficient quantities; shade and calculate DWL; and explain why the market trades too few lanterns.

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C2.2 | A Subsidy for Lanterns

To brighten the village, the Ministry pays sellers a subsidy of 6 coins per Lumos Lantern sold. For this question, ignore the external benefit: demand is the marginal benefit (MB) of a lantern.

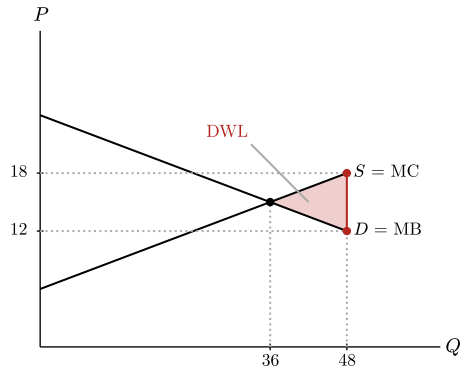
Buyer's Price (P_b): _____

Seller's Price (P_s): _____

Subsidy Quantity (Q^{Sub}): _____

Deadweight Loss (DWL): _____

Last Lantern Traded, MC vs. MB: ☐ above ☐ below



Answers: buyer's price **12 coins**; seller's price **18 coins**; subsidy quantity **48 lanterns**; DWL **36 coins**; MC is **above** MB for the last lantern traded.

Ignoring the external benefit, the unsubsidized market is efficient at $Q = 36$ and $P = 15$. The subsidy makes the seller's price 6 coins higher than the buyer's:

$$P_s = P_b + 6 \Rightarrow 6 + \frac{Q}{4} = 24 - \frac{Q}{4} + 6 \Rightarrow Q^{Sub} = 48.$$

$$P_b = 24 - \frac{48}{4} = 12, \qquad P_s = 6 + \frac{48}{4} = 18.$$

Check: $18 - 12 = 6$. At the last trade, MC is 18 and MB is 12. The added trades between 36 and 48 cost more than their private benefit, giving:

$$\text{DWL} = \frac{1}{2} \times 6 \times (48 - 36) = 36 \text{ coins}.$$

Pass (C2.2): Use the subsidy wedge to find both prices and quantity, identify MC above MB, and calculate DWL using the no-externality baseline of 36, rather than the 44 from C1.2.

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C3.1 | Designing the Policy

The Ministry of Magic has hired you to design a policy that corrects the market failure caused by the positive externality of Lumos Lanterns. Propose the policy you recommend and how it would impact the market for lanterns and their neighbors. Count the external benefit again, and replace the earlier policy rather than adding to it.

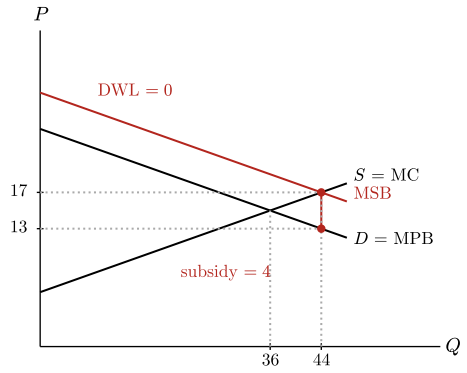
Policy Type: _____

Policy Size: _____

Post-Policy Quantity (Q^{Policy}): _____

Post-Policy Deadweight Loss: _____

Post-Policy Quantity vs. Efficient: ☐ above ☐ at ☐ below



Answers: a per-unit **subsidy** of **4 coins per lantern**; post-policy quantity **44 lanterns**; remaining DWL **0 coins**; quantity is **at** the efficient level.

Count the external benefit again. Replace the earlier 6-coin subsidy with a subsidy equal to the 4-coin external benefit. This is a total subsidy of 4 coins, not an additional payment on top of the earlier policy.

The subsidy makes the seller's price 4 coins higher than the buyer's:

$$P_s = P_b + 4 \Rightarrow 6 + \frac{Q}{4} = 24 - \frac{Q}{4} + 4 \Rightarrow Q^{Policy} = 44.$$

$$P_b = 24 - \frac{44}{4} = 13, \qquad P_s = 6 + \frac{44}{4} = 17.$$

At 44 lanterns, MSB equals MC: $28 - 44/4 = 6 + 44/4 = 17$. The policy reaches the efficient quantity from C1.2 and leaves no deadweight loss when the external benefit is counted.

Relative to the unregulated market, the corrective subsidy increases purchases from 36 to 44 lanterns. Buyers pay less, sellers receive more, and neighbors benefit from the additional light. Relative to the earlier 6-coin subsidy, the corrective policy lowers the payment and reduces quantity from 48 to 44, eliminating overproduction. These comparisons explain the effect; they are not extra required answer blanks.

Pass (C3.1): Choose a subsidy of 4, calculate its quantity, identify that quantity as efficient, and give zero remaining DWL. Use the positive externality to justify the subsidy. Assess this skill independently of arithmetic slips in other skills.